

MERCEL VUBANGSI

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OBJECTIVE

Machine Learning Engineer and Data Scientist with over 7 years of extensive and multifaceted experience in meticulously designing, diligently training, and expertly deploying highly advanced machine learning models to address intricate and multifaceted challenges spanning diverse domains such as healthcare, education, materials science, and IoT applications. Exceptionally proficient in Python, TensorFlow, PyTorch, and modern cloud ecosystems including AWS and GCP, demonstrating remarkable strengths in predictive modeling, deep learning methodologies, and sophisticated optimization strategies. Highly adept at orchestrating comprehensive and streamlined end-to-end ML workflows—encompassing data ingestion, feature engineering, model development, hyperparameter tuning, and production deployment—consistently delivering scalable, impactful solutions that significantly enhance operational efficiency and facilitate informed decision-making processes across various contexts. Complemented by a robust and well-honed quality assurance expertise, ensuring system reliability and performance through rigorous testing protocols, meticulous validation procedures, and continuous improvement efforts tailored to project needs. Deeply passionate about leveraging cutting-edge data-driven innovation to solve pressing real-world problems while steadfastly upholding an unwavering commitment to precision, technical excellence, and impactful outcomes.

EDUCATION

Master of Science in Computer Science, Maharishi International University Expected 2024 - 2027
Relevant Coursework: Machine Learning, Enterprise Architecture, Algorithms, Cloud Computing, Big Data Analytics

Master of Science in Artificial Intelligence Engineering, Near East University 2022 - 2023
GPA: 3.75/4.0
Thesis: "Optimizing Predictive Models for IoT-Driven Healthcare Systems Using Ensemble Learning"
Relevant Coursework: AI and Blockchain, AI and Cloud Computing, AI and IoT, Computational Intelligence

Bachelor of Technology in Software Engineering, HIBUMS Polytechnic 2018 - 2020
GPA: 3.8/4.0
Capstone Project: "Real-Time Resource Allocation System Using Linear Programming and Machine Learning"

PhD in Computational Physics, The University of Dschang, Cameroon 2013 - 2017
Grade: Honors
Dissertation: "Computational Modeling of Quantum Transport in Nanostructures Using ML-Enhanced Simulations"

MSc. in Computational Physics, The University of Dschang, Cameroon 2009 - 2012
GPA: 3.7/4.0
Thesis: "Effect of White Noise on Coherence of Levitating Electrons as Quantum Bits"

Bachelor of Science in Physics, University of Yaounde I 2002 - 2005
GPA: 14/20

SKILLS

ML & Data Science	Python, TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, Keras, Pandas, NumPy, SciPy, NLTK, SpaCy, Hugging Face Transformers, OpenCV
Programming	SQL, MATLAB, C++, Java, ReactJS, Node.js, Flask, Django, RESTful APIs, Shell Scripting
Cloud & DevOps	AWS (SageMaker, Lambda, Kinesis, EC2, S3), GCP (BigQuery, AI Platform), Docker, Kubernetes, Jenkins, Terraform
Data Engineering	Apache Spark, Apache Kafka, Hadoop, Snowflake, Elasticsearch, MySQL, PostgreSQL, MongoDB
Visualization	Tableau, Power BI, D3.js, Matplotlib, Seaborn, Plotly, Dash
QA & Optimization	Linear Programming (PuLP, Gurobi), PyTest, Jest, A/B Testing, Statistical Validation, Performance Benchmarking
Soft Skills	Project Management, Collaboration, Technical Writing, Problem-Solving, Leadership

EXPERIENCE

ML Engineer / Data Scientist

Jul 2022 - August 2025

International Research Center for AI and IoT

Nicosia, North Cyprus

- Architected a comprehensive and innovative healthcare chatbot utilizing TensorFlow and AWS SageMaker, incorporating LSTMs and attention mechanisms enhanced by transfer learning, which successfully reduced patient response times by 20% while markedly improving the overall quality of user interactions and satisfaction levels.
- Developed an advanced and highly effective IoT-driven predictive maintenance system with PyTorch and AWS Lambda, leveraging GRU models alongside sophisticated time-series forecasting techniques to accurately predict equipment failures well in advance, thereby minimizing downtime and enhancing operational continuity effectively.
- Constructed a robust and efficient real-time anomaly detection engine utilizing AWS Kinesis and XGBoost, meticulously optimizing detection thresholds with PuLP and SciPy to achieve exceptional accuracy and substantially reduce the frequency of false positives in critical operational monitoring scenarios.
- Designed and implemented a highly sophisticated ensemble learning pipeline featuring Random Forest and LightGBM for patient readmission prediction, employing SHAP for detailed interpretability and advanced feature engineering to significantly enhance predictive outcomes across diverse clinical contexts.
- Engineered an innovative and forward-thinking generative AI model utilizing GANs to simulate a wide range of patient health scenarios, thereby providing valuable and actionable support for clinical trial simulations with notably faster turnaround times compared to conventional methodologies.
- Utilized PyTest and A/B testing frameworks for thorough and meticulous quality assurance processes, enhancing overall model reliability by 8% and ensuring strict compliance with HIPAA regulations through carefully designed and systematically implemented audit trails.
- Analyzed intricate and multifaceted patient behavior trends with highly optimized BigQuery SQL queries, enabling seamless and efficient real-time analytics, and constructed visually rich Tableau dashboards enriched with Plotly visualizations to effectively track critical KPIs for operational insights and strategic planning.

Computational Scientist / ML Engineer

Dec 2019 - Mar 2022

Computational Materials Science Laboratory

Bambili, Cameroon

- Developed a sophisticated and meticulously crafted Scikit-learn ML pipeline incorporating ensemble methods like Random Forest and Gradient Boosting to predict material properties accurately, improving predictive accuracy by 10% through the strategic application of PCA techniques and dimensionality reduction.
- Engineered a high-performance and technically advanced molecular dynamics engine utilizing C++ and MPI, seamlessly integrating TensorFlow-based surrogate models to substantially accelerate simulation processes while maintaining exceptional computational integrity and precision.
- Constructed an interactive and user-friendly real-time visualization platform using Flask and D3.js, embedding machine learning predictions to enhance accessibility and usability for researchers exploring complex and multifaceted simulation results in detail.

- Designed an efficient and highly optimized LP-based scheduling system with PuLP and Gurobi, optimizing lecture hall allocation across numerous departments and successfully reducing scheduling conflicts by 25% through dynamic constraint management and robust operational planning.
- Employed PyTest for comprehensive and rigorous quality assurance on simulation outputs, ensuring high reproducibility and reliability with thorough statistical validation against a broad range of established experimental benchmarks.
- Optimized intricate and resource-intensive data pipelines leveraging Apache Spark and Hive, enabling highly scalable and efficient analysis of material properties under a variety of challenging experimental conditions and scenarios.

Computational Scientist

Jul 2017 - Dec 2019

Condensed Matter, Electronics, and Signal Processing Laboratory

Dschang, Cameroon

- Developed a TensorFlow-based CNN classifier for detailed spectroscopy data analysis, automating feature extraction processes with cutting-edge techniques to achieve exceptional accuracy in identifying intricate spectral patterns efficiently and reliably.
- Optimized quantum transport simulations utilizing Python and Quantum Espresso, incorporating kernel ridge regression techniques to enhance computational efficiency and predictive reliability across a wide array of simulations conducted over multiple iterations.
- Enhanced signal processing algorithms within MATLAB, meticulously refactoring FFT-based methods with custom-designed filters to significantly reduce analysis time by 30% while improving resolution for challenging and noisy low-signal environments.
- Automated critical and essential data workflows with LabVIEW and PostgreSQL, employing PyTest for thorough quality assurance to ensure robust and unwavering data integrity under demanding high-load operational conditions encountered regularly.
- Collaborated closely and productively with experimental physicists to validate machine learning models against detailed laboratory results, refining hyperparameters with grid search to improve prediction consistency and reliability across various experimental setups.
- Created an XGBoost-based predictive model to forecast simulation convergence with high accuracy, allowing preemptive and proactive parameter adjustments to streamline computational workflows and enhance overall efficiency.

Full Stack Engineer (Freelance)

Mar 2022 - Jul 2022

Cyprus Massachusetts Center for Innovation

Nicosia, North Cyprus

- Developed a comprehensive and fully functional school management system utilizing React and Node.js, integrating a Scikit-learn K-means algorithm to optimize classroom allocation effectively, reducing scheduling conflicts by 20% across multiple educational facilities and departments.
- Constructed robust and highly efficient RESTful APIs with Express and MongoDB, optimizing query performance through advanced indexing and caching techniques to ensure seamless data handling and exceptionally rapid response times for users.
- Secured the entire platform with sophisticated JWT and OAuth2 authentication protocols, utilizing Jest for meticulous quality assurance to meet stringent OWASP security standards without encountering any critical vulnerabilities during testing phases.
- Designed an interactive and visually appealing D3.js dashboard incorporating WebSockets for real-time updates on scheduling metrics, significantly enhancing administrative decision-making capabilities and streamlining operational workflows across the institution.
- Automated the deployment pipeline with GitHub Actions, ensuring comprehensive and reliable test coverage with Jest to maintain high system reliability and streamline release cycles efficiently throughout the development process.

Research Assistant (Computational Physics)

Jan 2010 - Jun 2013

University of Dschang Physics Department

Dschang, Cameroon

- Simulated quantum dot behavior under varying electric fields with MATLAB, improving overall model accuracy by 15% compared to traditional analytical approaches previously employed in computational physics research efforts.
- Constructed a rudimentary yet highly promising machine learning model utilizing Python and NumPy to predict material conductivity with notable accuracy, laying critical foundational groundwork for subsequent TensorFlow-based projects undertaken later.
- Optimized signal processing pipelines within Mathematica, achieving faster computation times through refined and carefully tuned algorithms validated against extensive and diverse laboratory measurements over multiple experimental runs.
- Employed statistical tools for thorough and detailed quality assurance on simulation outputs, ensuring a consistently high level of alignment and reliability with experimental data across a broad spectrum of trials and conditions.

PROJECTS

Diet Recommendation System Constructed a sophisticated Python-based diet planner utilizing Scikit-learn with K-means clustering and collaborative filtering techniques, improving user satisfaction by 85% through highly personalized and tailored recommendations; validated rigorously with PyTest to ensure consistency.

Financial Social Network App Developed a robust Java-based fundraising platform incorporating an XGBoost model for advanced risk assessment, optimizing savings allocation and operational efficiency through precise predictive analytics and strategic modeling.

AI-Enabled Digital Campus System Built an intelligent and adaptive campus management system with PyTorch, leveraging DQN reinforcement learning to optimize scheduling and resource allocation comprehensively, reducing operational costs by 18%; validated with rigorous statistical methods.

Fraud Detection System Engineered a LightGBM-based anomaly detection system utilizing Apache Spark, accurately identifying fraudulent patterns with exceptional precision; tested thoroughly with A/B testing frameworks to ensure reliability.

Medical Imaging Classifier Created a TensorFlow CNN integrated with OpenCV for meticulous X-ray image classification, accelerating diagnostic processes significantly with high accuracy; validated using advanced cross-validation techniques for robustness.

Quantum Material Simulator Developed a TensorFlow model to predict quantum transport properties efficiently with cutting-edge techniques, enhancing simulation performance through transfer learning; validated against detailed experimental benchmarks.

CERTIFICATIONS

Materials Data Sciences and Informatics – Georgia Institute of Technology, Coursera ([Verified](#))

REFERENCES

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Data Science / ML Lead

International Research Center for AI and IoT, Near East University

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Chair of Physics Department

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